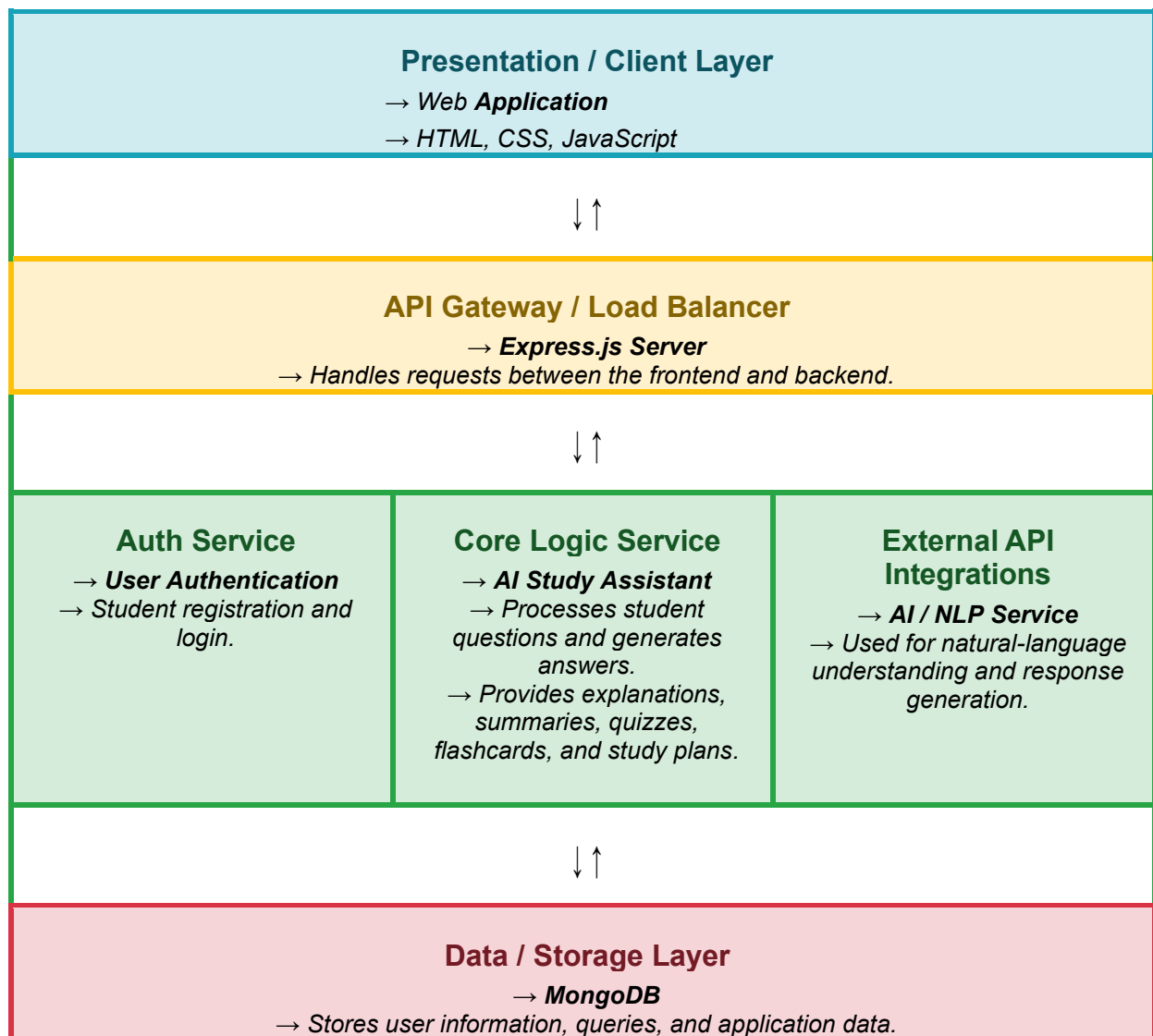


Solution Architecture

Date	28 September 2026
Team ID	SWTID-2026-7718
Project Name	AI Study Buddy API
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the interface through which students interact with the AI Study Buddy.	HTML, CSS, JavaScript
API Gateway	Receives requests from the frontend and routes them to the backend services.	Node.js, Express.js
Auth Service	Handles student registration and login.	Node.js, Express.js
Core Logic Service	Processes student queries and provides AI-based learning assistance.	AI, NLP, Node.js
External API	Provides AI-based natural-language processing and response generation.	AI/NLP API
Database	Stores user and application information.	MongoDB